

Program-6:Deploy an automated build pipeline using Docker Hub

Step 1: Create Dockerfile

nano Dockerfile

```
GNU nano 7.2 Dockerfile
FROM node:14
WORKDIR /app
COPY package*.json .
RUN npm install
COPY . .
EXPOSE 3000
CMD ["node", "app.js"]
```

Step 2: Build Docker Image

docker build -t program-6 .

Step 3: Tag Image for Docker Hub

docker tag program-6 yashwanth142003/program-6:latest

```
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~$ cd devOps
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps$ mkdir program6
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps$ cd program6
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ nano Dockerfile
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ docker build -t program-6 .
[+] Building 3.5s (11/11) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 144B 0.0s
=> [internal] load metadata for docker.io/library/node:14 3.4s
=> [auth] library/node:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 144B 0.0s
=> [1/5] FROM docker.io/library/node:14@sha256:a158d3b9b4e3fa813fa6c8c59 0.0s
=> CACHED [2/5] WORKDIR /app 0.0s
=> CACHED [3/5] COPY package*.json . 0.0s
=> CACHED [4/5] RUN npm install 0.0s
=> CACHED [5/5] COPY . . 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:c9d43c3a9ca5aeb3613aadd2dd1befb57ab4af164a7b0 0.0s
=> => naming to docker.io/library/program-6 0.0s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ docker tag program-6 yashwanth142003/program-6:latest
```

Step 4: Login to Docker Hub

docker login -u yashwanth142003

Step 5: Push Image to Docker Hub

docker push yashwanth142003/program-6:latest

```
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ docker login -u yashwanth142003
Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:
Login Succeeded
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ docker push yashwanth142003/program-6:latest
The push refers to repository [docker.io/yashwanth142003/program-6]
d4ceb38b206f: Pushed
70875a212c06: Pushed
5f70bf18a086: Pushed
681170253bcd: Pushed
0d5f5a015e5d: Pushed
3c777d951de2: Pushed
f8a91dd5fc84: Pushed
cb81227abde5: Pushed
e01a454893a9: Pushed
c45660adde37: Pushed
fe0fb3ab4a0f: Pushed
f1186e5061f2: Pushed
b2dba7477754: Pushed
latest: digest: sha256:2e1951c53f9b70d8d287430c7cee801dc24d40ab0a503b25910f2939c0e8fd98 size: 3041
```

The screenshot shows the Docker Hub interface for the repository `yashwanth142003/program-6`. The page includes a sidebar with navigation options like `Repositories`, `Hardened Images`, `Collaborations`, `Settings`, `Default privacy`, `Notifications`, `Billing`, `Usage`, `Pulls`, and `Storage`. The main content area displays the repository details, including the tag `latest` which was pushed 5 minutes ago. A table lists the image details: Digest `2e1951c53f9b`, OS/ARCH `linux/amd64`, Last pull `less than 1 day`, and Compressed size `333.38 MB`. A `Docker commands` section provides the command `docker pull yashwanth142003/program-6:latest` with a play button icon.

Step 6: Tag Image for GitHub Container Registry (GHCR)

docker tag program-6 ghcr.io/yashwanth142003/program-6:latest

Step 7: Login to GitHub Container Registry

```
echo ghp_<YOUR_GITHUB_TOKEN> | docker login ghcr.io -u  
yashwanth142003 --password-stdin
```

Step 8: Push Image to GHCR

```
docker push ghcr.io/yashwanth142003/program-6:latest
```

Step 9: Initialize Git Repository

```
git init
```

```
1rv24nc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ echo ghp_j3l0wyLpCQQBBTjyJdcf6spjduKx893YKwV2 | docker login ghcr.io -u yashwanth142003 --password-stdin  
WARNING! Your credentials are stored unencrypted in '/home/1rv24nc120_yashwanth_c/.docker/config.json'.  
Configure a credential helper to remove this warning. See  
https://docs.docker.com/go/credential-store/  
Login Succeeded  
1rv24nc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ docker push ghcr.io/yashwanth142003/program-6:latest  
The push refers to repository [ghcr.io/yashwanth142003/program-6]  
d4ceb38b206f: Pushed  
70875a212c06: Pushed  
5f70bf18a086: Layer already exists  
681170253bcd: Pushed  
0d5f5a015e5d: Layer already exists  
3c777d951de2: Layer already exists  
f8a91dd5fc84: Layer already exists  
cb81227abde5: Layer already exists  
e01a454893a9: Layer already exists  
c45660adde37: Layer already exists  
fe0fb3ab4a0f: Layer already exists  
f1186e5061f2: Layer already exists  
b2dba7477754: Layer already exists  
latest: digest: sha256:2e1951c53f9b70d8d287430c7cee801dc24d40ab0a503b25910f2939c0e0fd98 size: 3041  
1rv24nc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ git remote add origin https://github.com/Yashwanth142003/program-6.git  
fatal: not a git repository (or any of the parent directories): .git  
1rv24nc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ git init  
hint: Using 'master' as the name for the initial branch. This default branch name  
hint: is subject to change. To configure the initial branch name to use in all  
hint: of your new repositories, which will suppress this warning, call:  
hint:  
hint:     git config --global init.defaultBranch <name>  
hint:  
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and  
hint: 'development'. The just-created branch can be renamed via this command:  
hint:  
hint:     git branch -m <name>  
Initialized empty Git repository in /home/1rv24nc120_yashwanth_c/devOps/program6/.git/
```

Step 10: Add and Commit Files

```
git add .  
git commit -m "Initial commit for program-6"
```

Step 11: Add GitHub Remote Repository

```
git remote add origin https://github.com/Yashwanth142003/program-6.git
```

Step 12: Rename Branch to Main

it branch -M main

Step 13: Push Code to GitHub

git push -u origin main

```
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ git add .
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ git commit -m "Initial commit for program-6"
[master (root-commit) ef3b360] Initial commit for program-6
1 file changed, 7 insertions(+)
create mode 100644 Dockerfile
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ git remote add origin https://github.com/Yashwanth142003/program-6.git
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ git remote -v
origin https://github.com/Yashwanth142003/program-6.git (fetch)
origin https://github.com/Yashwanth142003/program-6.git (push)
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ git branch -M main
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program6$ git push -u origin main
Username for 'https://github.com': Yashwanth142003
Password for 'https://Yashwanth142003@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 329 bytes | 329.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Yashwanth142003/program-6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```



